Artificial Intelligence for Autonomous Systems

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Artificial Intelligence for Autonomous Systems (A31641)

Short Title:	AI Auton Syst		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Automotive, Automation and IoT		
Author	POLEARY		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

This module will build on the student s prior knowledge of computer algorithms and statistics to provide the student with a thorough understanding of many of the latest advancements in AI and Autonomous Systems. The main areas covered in this module are End to End Deep Learning, Computer Vision, Reinforcement Learning and applications of autonomous systems in real world settings.

Programmes

`COMP-0981`	Postgraduate Programmes in Computing and Mathematics (`WD_KGRAD_XX`)	5 / 0 / E

Indicative Content

- In depth treatment (assuming prior knowledge) of Deep Learning approaches.
- Convolutional Neural Networks
- Recurrent Neural Networks
- Long Term Short Term and Variational Auto Encoders
- Q-Learning and Policy Gradients
- Image Classification, Semantic Segmentation, Object Detection, Transfer Learning
- AI Toolkits and Platforms
- Multi-Input, Multi-Output Neural Networks.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate how the latest technologies in deep learning (techniques, architectures) operate and how they can be deployed.
2. Compare and contrast major deep learning approaches and assess their suitability for different scenarios and user requirements.
3. Design and implement solutions for autonomous systems.
4. Evaluate AI model training performance and fine tuning.
5. Assess the key aspects of an autonomous system including perception and action control loops.
6. Carry out an assessment of applying autonomous concepts to a system.
7. Design and implement a fully autonomous system using AI.

Learning and Teaching Methods

- Combination of lectures and computer-based practical and simulation exercises.
- The practical component will be delivered in one double lab session.
- Self-directed learning

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6,7
Assignment	50%	1,2,3,4,5,6,7
Lab Report	50%	3,4,6,7

Assessment Criteria

<40%: Unable to interpret and describe key concepts of the specific knowledge domains of AI, Deep Learning and Autonomous Systems and emerging technologies in these areas.

40%-59%: Able to interpret, describe and discuss key concepts of the specific knowledge domains of AI, Deep Learning and Autonomous Systems and emerging technologies in these areas. Able to discover and integrate related knowledge into application architectures.

60%-69%: Able to solve problems within the specific knowledge domains by experimenting with the appropriate skills and tools.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	24	
Independent Learning	222	

Essential Material

- "Tensorflow." https://www.tensorflow.org
- "keras.io." https://keras.io

Supplementary Material

- Gulli, A. and S. Pal. _Deep learning with Keras_. New York: Packt Publishing Ltd, 2017.
- Skansi, S. _Convolutional Neural Networks. In: Introduction to Deep Learning. Undergraduate Topics in Computer Science_. New York: Springer, 2018.
- Zafar, I., G. Tzanidou, R. Burton, N. Patel and L. Araujo. _Hands-on convolutional neural networks with TensorFlow: Solve computer vision problems with modeling in TensorFlow and Python_. New York: Packt Publishing Ltd, 2018.

Requested Resources

- COMPUTER LAB: BYOD Lab